

Slutsky Decomposition: A Clear Explanation

ECO 310 Midterm Review

The Big Picture

When the price of a good changes, your consumption of that good changes. But **WHY** does it change? The Slutsky decomposition shows that there are **TWO DISTINCT REASONS**:

The Two Effects

1. **Substitution Effect**: The good became relatively more/less expensive compared to other goods, so you substitute toward the relatively cheaper good
2. **Income Effect**: Your real purchasing power changed (you're effectively richer or poorer), so you consume more/less of the good

1 The Key Insight

Imagine the price of coffee rises from \$2 to \$4 per cup, and you reduce your consumption from 10 cups to 4 cups per week. The total change is -6 cups. But this -6 comes from:

- **Substitution**: Coffee is now twice as expensive relative to tea. Even if I gave you enough money to maintain your original satisfaction level, you'd still buy less coffee and more tea. Say you'd buy 7 cups instead of 10. That's -3 cups due to substitution.
- **Income effect**: The price increase makes you effectively poorer. With your original budget, you can't afford as much. This poverty effect makes you buy even less coffee, from 7 cups down to 4 cups. That's another -3 cups due to the income effect.

$$\text{Total change} = \text{Sub effect} + \text{Income effect} = (-3) + (-3) = -6$$

2 The Formula

Slutsky Decomposition

When price p_1 changes from p_1 to p'_1 :

$$\text{Total Effect} = \Delta x_1^{\text{total}} = x_1^{\text{new}} - x_1^{\text{original}}$$

This equals:

$$\begin{aligned} \text{Substitution Effect} &= x_1^{\text{compensated}} - x_1^{\text{original}} \\ + \text{Income Effect} &= x_1^{\text{new}} - x_1^{\text{compensated}} \end{aligned}$$

3 Step-by-Step Recipe

Step 1: Find the Original Bundle

At the initial prices (p_1, p_2) and income m , solve the consumer's utility maximization problem:

$$\begin{aligned} \max_{x_1, x_2} \quad & u(x_1, x_2) \\ \text{s.t.} \quad & p_1 x_1 + p_2 x_2 = m \end{aligned}$$

This gives you the **original bundle** (x_1^*, x_2^*) and **original utility** $\bar{u} = u(x_1^*, x_2^*)$.

Step 2: Find the Compensated Bundle

The price changes to (p'_1, p_2) . Now imagine we *compensate* the consumer by giving them just enough extra money so they can still *afford* their original utility $\bar{u}$ at the new prices.

Solve the expenditure minimization problem:

$$\begin{aligned} \min_{x_1, x_2} \quad & p'_1 x_1 + p_2 x_2 \\ \text{s.t.} \quad & u(x_1, x_2) = \bar{u} \end{aligned}$$

Or equivalently, find the bundle on the original indifference curve where $MRS = \frac{p'_1}{p_2}$.

This gives you the **compensated bundle** (x_1^c, x_2^c) .

Step 3: Find the New Bundle

Now solve the regular utility maximization at the new prices with the *original* income m :

$$\begin{aligned} \max_{x_1, x_2} \quad & u(x_1, x_2) \\ \text{s.t.} \quad & p'_1 x_1 + p_2 x_2 = m \end{aligned}$$

This gives you the **new bundle** $(x_1^{\text{new}}, x_2^{\text{new}})$.

Step 4: Calculate the Effects

Substitution Effect:

$$x_1^{\text{SE}} = x_1^c - x_1^*$$

Income Effect:

$$x_1^{\text{IE}} = x_1^{\text{new}} - x_1^c$$

Total Effect:

$$x_1^{\text{TE}} = x_1^{\text{new}} - x_1^* = x_1^{\text{SE}} + x_1^{\text{IE}}$$

4 Key Properties

1. Substitution Effect Sign:

- Always moves in the **opposite direction** of the price change
- If p_1 rises: $x_1^{\text{SE}} \leq 0$ (buy less)
- If p_1 falls: $x_1^{\text{SE}} \geq 0$ (buy more)

2. Income Effect Sign:

- Depends on whether the good is **normal** or **inferior**
- **Normal good:** When effectively poorer, buy less
 - If p_1 rises (poorer): $x_1^{\text{IE}} < 0$
 - If p_1 falls (richer): $x_1^{\text{IE}} > 0$
- **Inferior good:** When effectively poorer, buy more
 - If p_1 rises (poorer): $x_1^{\text{IE}} > 0$
 - If p_1 falls (richer): $x_1^{\text{IE}} < 0$

3. Total Effect: Sum of the two effects. For normal goods:

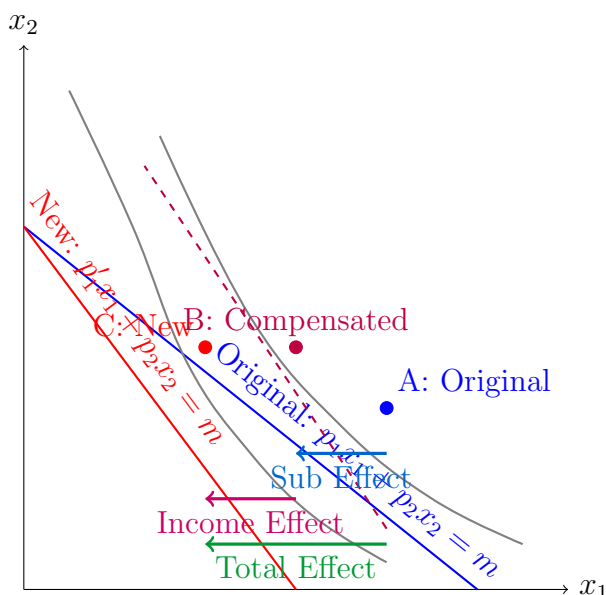
- Both effects work in same direction
- Price rises $\Rightarrow$ quantity demanded falls

4. Giffen Good: An inferior good where the income effect dominates

- $|x_1^{\text{IE}}| > |x_1^{\text{SE}}|$
- Price rises but quantity demanded rises too!
- Very rare in practice

5 Graphical Interpretation

Consider a price increase for good 1:



Interpretation:

- **Point A → Point B (Substitution Effect):** Move along the original indifference curve to the new price ratio. The consumer substitutes away from the now-relatively-expensive good 1.
- **Point B → Point C (Income Effect):** Remove the compensation. The consumer moves to a lower indifference curve because they're effectively poorer. For a normal good, they buy less of both goods.
- **Point A → Point C (Total Effect):** The observed change in consumption.

6 Special Cases

Perfect Complements: $u = \min\{ax_1, bx_2\}$

Substitution Effect = 0

With perfect complements, you always consume at the kink where $ax_1 = bx_2$. When the price changes, the compensated bundle has the exact same quantities as the original bundle (just on a different budget line). There's no substitution!

Income Effect = Total Effect

The entire change in consumption is due to the income effect.

Quasilinear: $u = v(x_1) + x_2$

Income Effect = 0 (for x_1 at interior solutions)

The optimal x_1 depends only on the price ratio, not on income. So the income effect for good 1 is zero!

Substitution Effect = Total Effect

The entire change in x_1 consumption is due to the substitution effect.

7 Concrete Example

Setup: $u = x_1x_2$, $m = 100$, $p_2 = 1$, p_1 rises from 1 to 2.

Step 1: Original Bundle

With $p_1 = 1, p_2 = 1, m = 100$:

$$\begin{aligned}x_1^* &= \frac{m}{2p_1} = \frac{100}{2} = 50 \\x_2^* &= \frac{m}{2p_2} = \frac{100}{2} = 50 \\ \bar{u} &= 50 \times 50 = 2500\end{aligned}$$

Step 2: Compensated Bundle

Need $x_1x_2 = 2500$ and $MRS = \frac{x_2}{x_1} = \frac{p'_1}{p_2} = \frac{2}{1} = 2$.

From $\frac{x_2}{x_1} = 2$, we get $x_2 = 2x_1$.

Substituting into $x_1x_2 = 2500$:

$$x_1 \cdot 2x_1 = 2500 \implies x_1^c = \sqrt{1250} \approx 35.36$$

$$x_2^c = 2x_1^c \approx 70.71$$

Step 3: New Bundle

With $p'_1 = 2, p_2 = 1, m = 100$:

$$\begin{aligned}x_1^{new} &= \frac{m}{2p'_1} = \frac{100}{4} = 25 \\x_2^{new} &= \frac{m}{2p_2} = \frac{100}{2} = 50\end{aligned}$$

Step 4: Calculate Effects

$$\text{Substitution Effect: } x_1^c - x_1^* = 35.36 - 50 = -14.64$$

$$\text{Income Effect: } x_1^{new} - x_1^c = 25 - 35.36 = -10.36$$

$$\text{Total Effect: } x_1^{new} - x_1^* = 25 - 50 = -25$$

Check: $-14.64 + (-10.36) = -25 \checkmark$

Interpretation: When coffee price doubled, you reduced consumption from 50 to 25 cups. About 59% of this reduction (-14.64) was due to substitution toward tea, and 41% (-10.36) was due to being effectively poorer.

8 Summary Checklist

Quick Reference

- **Three bundles:** Original, Compensated, New
- **Compensated bundle:** Stay on original IC, move to new price ratio
- **Sub effect:** Always opposes price change (≤ 0 if price rises)
- **Income effect:** Depends on normal/inferior
- **Normal good:** Both effects same direction when price changes
- **Perfect complements:** Sub effect = 0
- **Quasilinear:** Income effect = 0 (for x_1)
- **Total = Sub + Income** always!